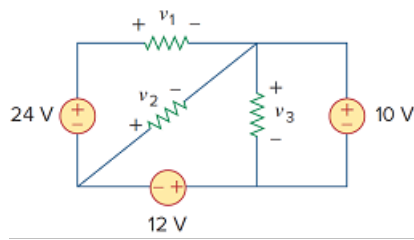


Problem

Obtain v_1 through v_3 in the circuit of Fig. 2.81.

Figure 2.81:



Step-by-step solution

Step 1 of 1

Applying KVL around the entire outside loop we get,

$$-24 + v_1 + 10 + 12 = 0 \text{ or } v_1 = \mathbf{2V}$$

Applying KVL around the loop containing v_2 , the 10-volt source, and the 12-volt source we get,

$$v_2 + 10 + 12 = 0 \text{ or } v_2 = \mathbf{-22V}$$

Applying KVL around the loop containing v_3 and the 10-volt source we get,

$$-v_3 + 10 = 0 \text{ or } v_3 = \mathbf{10V}$$